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Dr. MURTAZA SAADIQUE BASHA
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Mrs . SRUTHI
Mr. GANGU RAMA NAIDU

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PREFACE

Welcome to a thrilling journey through the world of Artificial Intelligence (AI) and Machine Learning (ML), two of the most transformative technologies of our time, and the versatile programming language that powers them – Python. This book is your entry point to understanding how Python has become the language of choice for AI and ML, and how these technologies are reshaping industries, enhancing decision-making, and fueling innovation. Whether you are a seasoned programmer, a data enthusiast, or someone just starting to explore the vast realm of AI and ML, this text promises to be an enlightening adventure. In the age of data, AI and ML have emerged as the driving forces behind some of the most remarkable advancements in science, business, healthcare, and beyond. Python, with its simplicity, versatility, and a wealth of libraries and tools, has become the preferred choice for those looking to harness the potential of AI and ML. The primary aim of this book is to provide you with a comprehensive understanding of AI and ML using Python. As you embark on this journey, you will learn about the core principles, algorithms, and real-world applications of these technologies. The authors have thoughtfully crafted this content to make it accessible to both experienced *professionals and those new to the world of data science. Our journey takes us through the fundamentals of AI and ML, delving into the essential mathematics, the Python programming required to build and deploy models, and the best practices for handling data. Whether it's predictive analytics, natural language processing, computer vision, or recommendation systems, we will explore how Python can be applied to a diverse range of AI and ML applications. Each chapter offers a balanced blend of technical insights and practical applications, ensuring that you can grasp the intricacies of AI and ML while appreciating their real-world impact. We will also explore the ethical considerations, best practices, and the potential challenges that come with using AI and ML in various domains. As you dive into the world of AI and ML with Python, we invite you to envision the possibilities that these technologies present. The advancements discussed in these pages are not just about building models but about how AI and ML are enhancing decision-making, automating processes, and driving innovation across industries.*

- Authors

AUTHOR BIOGRAPHY

Dr. Murtaza Saadique Basha is working as Professor and Head, Department of Master of Computer Applications, C.Abdul Hakeem College of Engineering & Technology, Melvisharam. He did his Masters in Computer Application at University of Madras, Chennai. He received his Master's in Computer Science and Engineering from Dr.M.G.R Educational and Research Institute, Chennai. He is the gold medalist with honours in his M.Tech course. He received his Doctorate of Philosophy from Manonmaniam Sundaranar University under Computer Science and Engineering discipline. He has also cleared State Eligibility Test (SET) conducted by Mother Terasa University. He is having more than 17 years of teaching experience.

Dr. Thiruma Valavan A is a multifaceted expert with a wealth of knowledge in a variety of sectors, including Marketing, Banking, Management, Compliance, Human Resource Management, Information Technology, Cyber Security, and Computer Programming. His focus areas are Emerging Technologies, like, Artificial Intelligence (AI) and Machine Learning (ML), Internet of Things (IoT), Virtual Reality and

Augmented Reality (VR and AR), Robotics and Automation, Blockchain Technology and 3D printing, and he believes that, these technologies are expected to have a significant impact on various industries and daily life, in the coming up days. He had more than 30 years of rich expertise in the banking and finance sector. He headed one of India's biggest banks, Canara Bank, with branches around the country in rural, semi-urban, and metropolitan areas, as well as senior positions at the regional, head office, and training setup. He is currently serving as Deputy Director of the Indian Institute of Banking and Finance. His credentials include the CAIIB, M Com, MA in Sociology, MCA, PCP in Blockchain Technology from IIT-Kanpur, Marketing Management from IIM-Bangalore, MBA in Marketing, MPhil in Entrepreneurship, and Ph.D. in Management. Additionally, he is a Certified Bank Trainer.

Other certificates of specialization include:

1. "Customer Relationship Management" from IIM-Bangalore.
2. "Data Science, Machine Learning" from Harvard University.
3. "Digital Competition in Financial Services" from Copenhagen Business School.
4. "Financial Development and Financial Inclusion" from International Monetary Fund (IMF) Institute
5. "Unlocking Investment and Finance in Emerging Markets and Developing Economies (EMDEs)" from World Bank Group.

More than 20+ Research papers have been published in SCIE/SCOPUS/Google-indexed international journals. He has attended and presented many research articles at National & International conferences.

Dr. H Parthasarathi has been working as an Associate Professor of the Department of Computer Science and Engineering at Gayatri Vidya parishad College of Engineering (A), Visakhapatnam since 5 years. He has about 17 years of teaching experience from different Engineering College and universities.

Dr. H Parthasarathi has currently guiding 4 research scholars from different universities and has published more than 10 research papers in referred journals, 2 Books and 4 patents granted in different country. He has completed his PhD in computer Science and Engineering from Birla Institute of Technology, Mesra, and Ranchi. Jharkhand, India. He received his ME (Software Engineering) Degree from Jadavpur University, West Bengal, India. His Research area is Software Engineering, Machine Learning, Natural Language Processing and Computer Vision.

Mrs. Sruthi is a young and dynamic individual with a strong academic background in computer science and engineering. She is currently working as an Assistant Professor in the Department of computer science and engineering, KSR College for engineering and technology, Tiruchengode .she is passionate about research, believes in learning, striving for enhancing skills.

Mr. Gangu Rama Naidu received M.Tech (VLSID) from Aditya Engineering College, Surampalem affiliated to JNTUK University, and pursuing Ph. D. Under the guidance of Dr.Ch.Srinivasa Rao from JNTUK University. I have 12 years of experience in teaching and administration. I am currently working as Associate Professor, Aditya Engineering College, Surampalem, India.

Contents

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Mrs. SRUTHI is a young and dynamic individual with a strong academic background in computer science and engineering. She is Currently working as an Assistant Professor in the Department of computer science and engineering, KSR college of engineering, Tiruchengode. She is passionate about research, believes in learning, striving for enhancing skills.

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